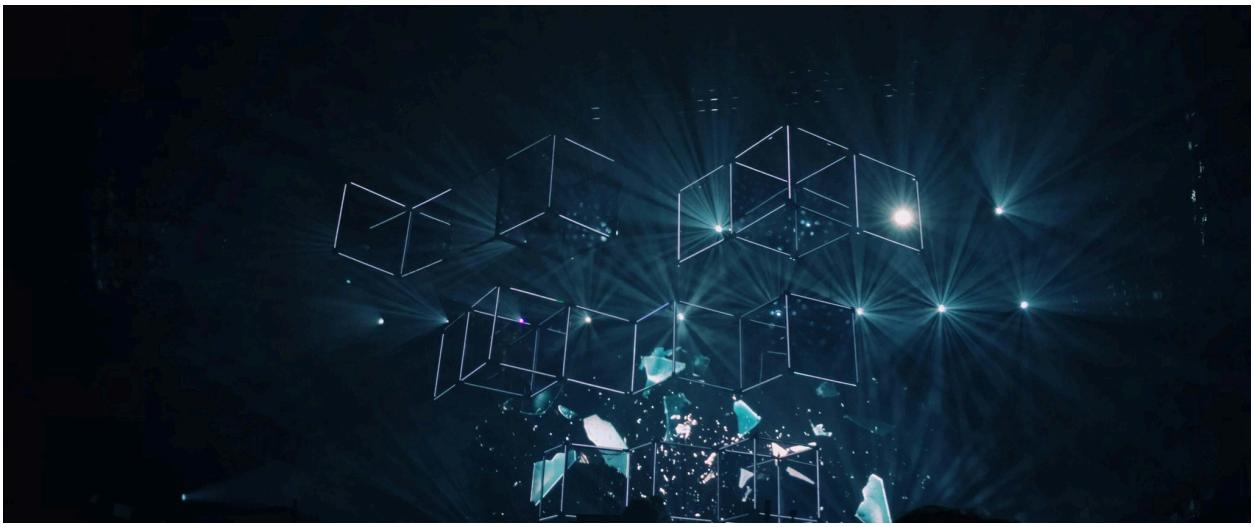

DCL and TCL in SQL

▼ DCL - Data Control Language

What it is

- Controls permissions and access to database objects
- Typical users: DBAs and system owners

Core commands

- **GRANT**: Give privileges to users or roles
- **REVOKE**: Remove previously granted privileges

Common privileges

- SELECT
- INSERT
- UPDATE
- DELETE

Examples

MySQL

```
-- Create user
CREATE USER 'alice'IDENTIFIED BY 'secret';

-- Grant read-only on a database
GRANT SELECT ON analytics.* TO 'alice';

-- Grant specific table writes
GRANT INSERT, UPDATE ON analytics.orders TO 'alice';

-- Revoke
REVOKE UPDATE ON analytics.orders FROM 'alice';
```

▼ TCL-Transaction Control Language

What it is

- Manages the lifecycle of transactions to ensure data integrity
- Typical users: application developers and analysts writing multi-step changes

Core commands

- BEGIN or START TRANSACTION
- COMMIT
- ROLLBACK
- SAVEPOINT name
- ROLLBACK TO SAVEPOINT name

Examples

Basic transaction with savepoints

```
BEGIN;  -- or START TRANSACTION

INSERT INTO orders(id, customer_id, amount) VALUES (1001, 42, 250.00);
```

```
SAVEPOINT before_discount;

UPDATE orders SET amount = amount * 0.9 WHERE id = 1001; -- apply 1
0% discount

-- Oops, wrong logic. Roll back just the discount step.
ROLLBACK TO SAVEPOINT before_discount;

-- Continue with correct logic
UPDATE orders SET amount = amount - 20 WHERE id = 1001;

COMMIT;
```